
Linear Algebra I-II

Theorems & Lemmas

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1 Vector Spaces

Definition 1.1: Vector Space

A **vector space** over a field K is a set V , endowed with two operations

$$\begin{aligned} + : V \times V &\rightarrow V, & (v_1, v_2) &\mapsto v_1 + v_2, \\ \cdot : K \times V &\rightarrow V, & (\alpha, v) &\mapsto \alpha \cdot v \end{aligned}$$

s.t. the following conditions (axioms) hold:

$$(V1) \quad \forall v_1, v_2, v_3 \in V : v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3,$$

$$(V2) \quad \exists \text{ an element } 0 \in V : 0 + v = v \quad \forall v \in V,$$

$$(V3) \quad \forall v \in V \exists v' \in V : v + v' = 0,$$

$$(V4) \quad \forall v_1, v_2 \in V : v_1 + v_2 = v_2 + v_1,$$

$$(V5) \quad \forall a, b \in K \forall v \in V : a \cdot (b \cdot v) = (a \cdot b) \cdot v,$$

$$(V6) \quad \forall v \in V : 1 \cdot v = v,$$

$$(V7) \quad \forall a \in K \forall v_1, v_2 \in V : a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2,$$

$$(V8) \quad \forall a, b \in K \forall v \in V : (a + b) \cdot v = a \cdot v + b \cdot v.$$

Lemma 1.2: Properties of Vector Spaces

Let V be a vector space over a field K . Then:

- (a) The 0 vector from (V2) is unique. To avoid confusion (e.g. with $0 \in K$) we will sometimes denote it by 0_V .
- (b) The element v' (corresponding to v) from (V3) is unique. We'll denote it by $-v$. We'll also write $w - v := w + (-v)$ for all $w \in V$,
- (c) $\forall v \in V : 0 \cdot v = 0$,
- (d) $\forall a \in K : a \cdot 0 = 0$,
- (e) $\forall v \in V : -1 \cdot v = -v$,
- (f) $\forall v \in V : -(-v) = v$,
- (g) If $a \cdot v = 0$ for some $a \in K, v \in V$, then either $a = 0$ or $v = 0$,
- (h) Associativity for $+$ holds also for some of n elements. $v_1 + \dots + v_n$ makes sense no matter how we put the brackets $v_1 + (v_2 + (\dots + (v_{n-1} + v_n)))$ or $(v_1 + v_2) + (\dots)$ etc.

Example: The Coordinate Space K^n

Let K be a field and $n \in \mathbb{N}$.

$$K^n = \{(a_1, \dots, a_n) \mid a_i \in K, 1 \leq i \leq n\}.$$

K^n is a vector space over K if we endow it with the following operations:

- Let $v = (a_1, \dots, a_n), w = (b_1, \dots, b_n) \in K^n$.

$$v + w = (a_1, \dots, a_n) + (b_1, \dots, b_n) := (a_1 + b_1, \dots, a_n + b_n),$$

- For $a \in K$

$$a \cdot v = a \cdot (a_1, \dots, a_n) := (a \cdot a_1, \dots, a \cdot a_n)$$

- $0 := (0, \dots, 0) \in K^n$

With these operations K^n becomes a vector space over K .

Definition 1.3: Linear Subspace

Let V be a vector space over K . A subset $W \subseteq V$ is called a **linear subspace** of V (or just subspace of V), if the following holds

(LSS1) $W \neq \emptyset$,

(LSS2) $\forall w_1, w_2 \in W : w_1 + w_2 \in W$,

(LSS3) $\forall a \in K, w \in W : a \cdot w \in W$.

Lemma 1.4: Linear Subspace is a Vector Space

Let V be a vector space over K and $W \subseteq V$ a subspace. Then W is a vector space on its own, when endowed with the operations from V .

Lemma 1.5: Criterion for a Linear Subspace

Let V be a vector space over K , and $W \subseteq V$ a subset. Then W is a linear subspace of V if and only if the following holds:

(1) $0_V \in W$,

(2) $\forall a_1, a_2 \in K, w_1, w_2 \in W : a_1 w_1 + a_2 w_2 \in W$.

Example: Spaces of functions

Fix a field K and a non-empty set S . Define

$$K^S = \{f : S \rightarrow K \mid f \text{ is a function}\}.$$

Define $+$ and $\cdot$ on K^S as follows. Let $f, g \in K^S, a \in K$. Define

$$(f + g) : S \rightarrow K, \quad S \ni x \xrightarrow{f+g} f(x) + g(x),$$

$$(a \cdot f) : S \rightarrow K, \quad S \ni x \xrightarrow{a \cdot f} a \cdot f(x).$$

Now Let $S = [0, 1]$ and take $K = \mathbb{R}$. Define $C(S) := \{f \in \mathbb{R}^S \mid f \text{ is continuous on } [0, 1]\} \subseteq \mathbb{R}^S$. Then $C(S) \subseteq \mathbb{R}^S$ is a linear subspace.